

Zhongchun Zhou

Ph.D. Candidate in Electronic and Computer Engineering

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RESEARCH INTERESTS

Computer Architecture • GPGPU • AI Infrastructure • HPC • EDA • IC

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Hong Kong

Ph.D. Candidate in Electronic and Computer Engineering

Aug. 2022 – Present

Advisor: Prof. Wei Zhang CGA: 4.06/4.3 Research Focus: LLM Accelerators, Memory Hierarchy

Relevant Coursework: Advanced Computer Architecture (A), Advanced Artificial Intelligence (A+), Advanced Deep Learning Architectures (A), Microelectronics Fabrication Technology (A), Advanced Algorithms (A)

Tsinghua University

Beijing, China

B.E. in Microelectronic Engineering

Aug. 2018 – Jun. 2022

GPA: 3.77/4.0 IELTS: 7.5

PUBLICATIONS

- [1] B. Cui*, **Z. Zhou***, et al., “TileBench: A Controlled Benchmark for Performance Evaluation and Bottleneck Diagnosis of Tile-Based Programming Models”, *The 5th Asia-Pacific Chapter of the Association for Computational Linguistics & the 15th International Joint Conference on Natural Language Processing (ACL)*, 2026. (*Equal contribution)
- [2] **Z. Zhou**, C. Lai, Y. Gu, W. Zhang, “DCO: Dynamic Cache Orchestration for LLM Accelerators through Predictive Management”, *IEEE Transactions on Computers*, 2026. (Accepted) [arXiv:2512.07312]
- [3] **Z. Zhou**, Y. Gu, C. Lai, Y. Wang, W. Zhang, “Sim-FA: A GPGPU Simulator Framework for Fine-Grained Asynchronous Pipeline Analysis”, *IEEE Transactions on Computers*, 2026. (Under review) [arXiv:2605.00555]
- [4] **Z. Zhou**, C. Lai, W. Zhang, “LLAMCAT: Optimizing Large Language Model Inference with Cache Arbitration and Throttling”, *Proceedings of the 54th International Conference on Parallel Processing (ICPP)*, 2025.
- [5] C. Lai*, **Z. Zhou***, A. Poptani, W. Zhang, “LCM: LLM-focused Hybrid SPM-cache Architecture with Cache Management for Multi-Core AI Accelerators”, *Proceedings of the International Conference on Supercomputing (ICS)*, 2024. (*Equal contribution)
- [6] Y. Du, Y. Hu, **Z. Zhou**, Z. Zhang, “High-Performance Sparse Linear Algebra on HBM-Equipped FPGAs Using HLS: A Case Study on SpMV”, *Proceedings of the 30th ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA)*, 2022.
- [7] C. Lai, Y. Wang, **Z. Zhou**, W. Zhang, R. Chang, H. Yao, “DFAC: Dataflow-Aided Intra-cluster Coalescer for AI Accelerators”, *Proceedings of the 32nd Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2027.

RESEARCH EXPERIENCE

Ph.D. Research: Dynamic Cache Orchestration & LLM Accelerator Memory Systems

HKUST

Advisor: Prof. Wei Zhang, Fellow, IEEE

Feb. 2023 – Present

- Designed the **Tensor Management Unit (TMU)**—a lightweight hardware structure integrated into the memory hierarchy that exposes tensor- and tile-level metadata to guide cache replacement, dead-block prediction, and dynamic bypassing. Implemented in Chisel HDL and synthesized in a 15 nm standard-cell library (area: 0.064 mm², frequency: 2 GHz).
- Developed **LCM** (ICS’24) and its substantial extension **DCO** (IEEE TC): a hybrid SPM-cache architecture with self-adaptive anti-thrashing, predictive dead-block policies, and coordinated bypassing. Achieved up to **1.90×** speedup over conventional LRU-managed LLC across diverse cache sizes and FlashAttention-based LLM workloads (including GQA).
- Proposed **DFAC** (ASP-DAC’27): A dataflow-aided intra-cluster coalescing architecture conveying compile-time reuse counts to hardware merge tables, eliminating round-by-round L2 thrashing and alleviating NoC bisection pressure (up to **2.08×** speedup on FlashAttention and **1.22×** on large GEMM).
- Proposed **LLAMCAT** (ICPP’25): MSHR- and load-balance-aware cache arbitration combined with a novel two-level dynamic multi-gear thread throttling policy tailored for LLM inference. Delivered 1.10–2.11× speedup under memory-bound and cache-size-constrained scenarios.

- Built and validated a cycle-accurate analytical model that accounts for realistic overlapping behaviors between computation and memory traffic, enabling reliable projection of policy benefits to large-scale production workloads.
- *Key outcomes:* Publications [2], [4], [5], and [7]. The framework demonstrates that a shared system-level cache augmented with lightweight tensor-aware orchestration is a practical, scalable alternative to deeply hierarchical software-managed scratchpads in future AI accelerators.

Sim-FA: Cycle-Accurate GPGPU Simulator for Asynchronous Pipeline Analysis

HKUST

Advisor: Prof. Wei Zhang, Fellow, IEEE

Oct. 2025 – Present

- Built the first academic cycle-accurate simulator (Sim-FA) supporting emerging NVIDIA Hopper features—TMA, WGMMMA, warp specialization, and fine-grained asynchronous barriers—for FlashAttention-3 kernels.
- Achieved mean absolute percentage error (MAPE) of **5.7%** (max 12.7%) against real H800 hardware across wide ranges of model sizes and sequence lengths.
- Developed a complementary theoretical framework that reveals why existing analytical roofline models significantly underestimate DRAM traffic under realistic hardware constraints; validated against NCU profiles on Blackwell GPUs.
- *Key outcome:* Publication [3]. The simulator and analysis framework provide essential tools for academic research on modern AI infrastructure and hardware–software co-design.

TileBench: Controlled Benchmark for Tile-Based GPU Programming Models

GMU

Advisor: Prof. Keren ZHOU

2026

- Designed and implemented **TileBench**, a controlled and reproducible benchmark suite with 45 representative AI operators. The benchmark provides Triton and cuTile implementations, standardized multi-dtype and input-size sweeps, default/autotuned configurations, roofline-based metrics, and profiling-guided diagnosis using NCU and SASS inspection.
- Conducted systematic evaluation on NVIDIA B200 GPUs demonstrating that performance gaps are workload-dependent: cuTile excels on Tensor-Core/TMA-friendly kernels with regular, reusable tile movement, while Triton achieves stronger aggregate performance and is more robust on irregular, streaming, and bandwidth-bound operators. Autotuning yields measurable gains (1.15–1.20× roofline utilization improvement) yet substantial headroom remains relative to hardware limits.
- Evaluated LLM-generated (GPT-5.5 and Claude Opus 4.7) Triton and cuTile kernels under an iterative refinement protocol, showing that Triton is consistently more token-efficient than cuTile.
- *Key outcome:* TileBench [1] fills a critical gap by enabling fair, apples-to-apples comparison of tile-centric DSLs and serves as a foundation for future tooling and LLM-assisted kernel generation research in AI infrastructure.

GraphLily: Graph Processing Overlay on FPGA

Cornell University

Research Assistant (Advisor: Prof. Zhiru Zhang)

Jun. 2021 – Sep. 2021

- Implemented the connected-components algorithm on the HLS-based GraphLily overlay; extended the hardware operator set to meet algorithmic requirements and benchmarked against the Xilinx Vitis sparse library. [6]

Tsinghua Ventus GPGPU

Tsinghua University

Research Assistant (Advisor: Prof. Hu He)

Jan. 2022 – Jun. 2022

- Designed a GPU thread-block scheduler (CTA Scheduler) incorporating state-of-the-art scheduling policies; implemented in Chisel HDL and used in the Ventus GPGPU mainline (open source: github.com/THU-DSP-LAB/ventus-gpgpu).

PROFESSIONAL EXPERIENCE

Alibaba T-Head Silicon — Data Center & Compute Software

Intern Chip Engineer

Jul. 2025 – Oct. 2025

- Analyzed production FlashAttention-3 kernels, focusing on warp specialization and ping-pong scheduling.
- Performed micro-benchmarking of WGMMMA and TMA instructions to characterize real hardware behavior and guide simulator modeling (directly informed Sim-FA development).

HONORS & AWARDS

- RedBird Ph.D. Recruitment Award, The Hong Kong University of Science and Technology 2022
- Academic Scholarship of the Department of Electronic Engineering, Tsinghua University 2019

SKILLS

Hardware:	Verilog HDL, Chisel, HLS (Vitis), RTL Synthesis (Design Compiler)
Software & Sim:	C/C++, Triton/Gluon, CUDA, NVIDIA Nsight, cycle-accurate simulators
Languages:	English (IELTS 7.5)